Clustering MLB Hitters off their Two-Strike Approaches

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GOAL & INTRO:

- **GOAL:** Utilizing Statcast batting data, cluster players based on their two-strike hitting ability
- Which players perform best when there's two strikes?
- Do MLB players actually take a two-strike approach?

METHODOLOGY:

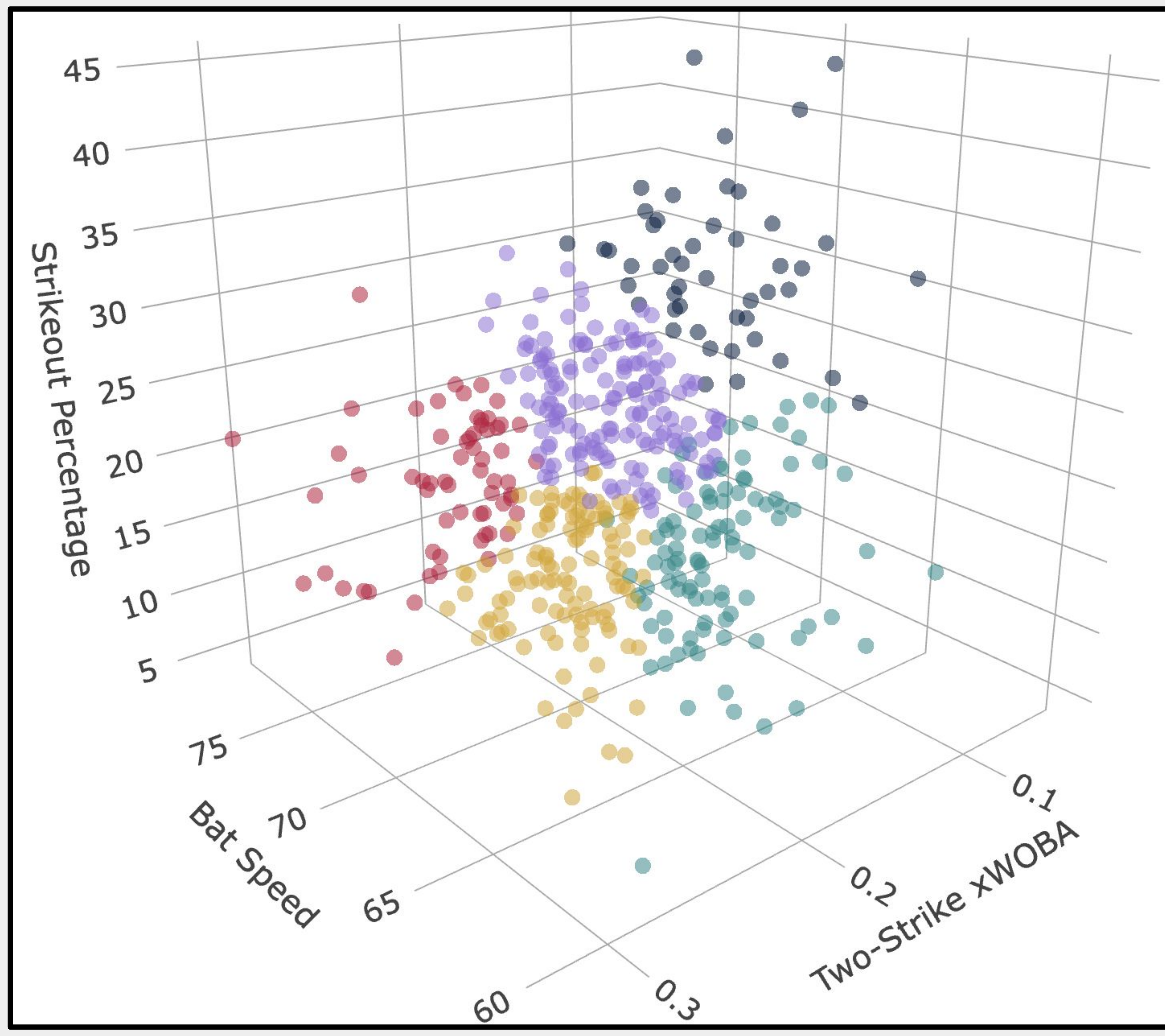
1. Data Collection and Cleaning
   a. Min. of 20 balls hit in play on two strikes in 2024 MLB season
2. Variable Selection
   a. Used:
      i. Bat Speed
      ii. Two-Strike xWOBA
      iii. Strikeout %
   b. Considered:
      i. Swing Length
      ii. Barrel %
3. K-Means Clustering

ANALYSIS & RESULTS:

- Cluster 1
  - Poor two-strike hitters
- Cluster 2
  - "Swing for the fences" approach
- Cluster 3
  - Majority takes contact approach
- Cluster 4
  - Similar to Cluster 3 with higher bat speed and lower strikeout %
- Cluster 5
  - Average MLB hitter on two strikes

FUTURE WORK:

- Limitations
  - Bat tracking data only available for the 2024 season
  - Bat speed and swing length derived from only balls with estimated WOBA values
- Future Work
  - Focus on individual players
  - General two-strike plan of attack for teams
  - Develop a metric that measures a player's ability on two strikes



Notable Players in Each Cluster:

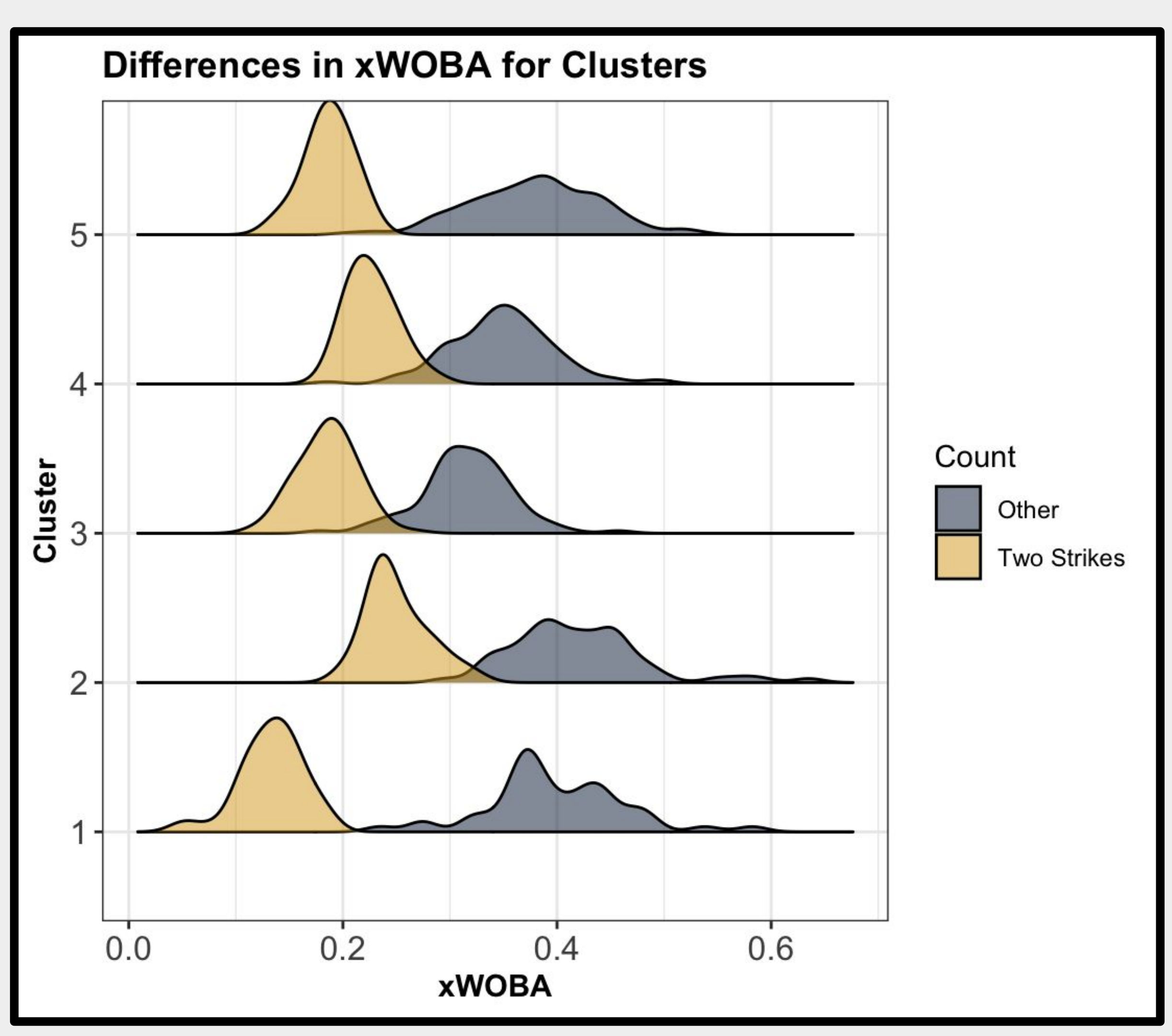
Cluster 1:
- Kris Bryant
- Henry Davis
- Austin Hedges
- Joey Gallo
- Luis Robert Jr.
- James Outman

Cluster 2:
- Aaron Judge
- Giancarlo Stanton
- Shohei Ohtani
- Mike Trout
- Fernando Tatis Jr.
- Bobby Witt Jr.

Cluster 3:
- Luis Arraez
- Jose Altuve
- Alex Verdugo
- Ozzie Albies
- Cody Bellinger
- Kevin Pillar

Cluster 4:
- Steven Kwan
- Trea Turner
- Christian Yelich
- Nolan Arenado
- Mookie Betts
- Freddie Freeman

Cluster 5:
- Oneil Cruz
- Javier Báez
- Elly De La Cruz
- Brent Rooker
- Jackson Holliday
- Jazz Chisolm Jr.



Cluster 1 struggles on two strikes, as seen by the difference in xWOBA

Differences in bat speeds show all clusters generally follow a two-strike approach

Median Statistical Measures per Cluster

CLUSTER	BAT SPEED (MPH)	TWO-STRIKE XWOBA	STRIKEOUT %	SWING LENGTH (INCHES)	BARREL %
1	69.22	0.136	31.73	7.25	5.00
2	71.74	0.242	20.16	7.44	9.60
3	64.87	0.188	17.08	6.94	2.17
4	68.23	0.224	15.45	7.18	4.29
5	69.18	0.187	24.17	7.31	6.88

Table provides median measures of each cluster to account for any skewness of the data